

Ensemble of Agents — Class 1

DSC 180A, Fall 2026 Domain D31

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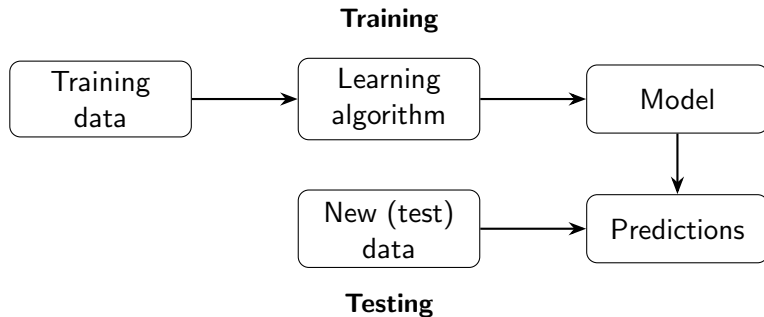
Administration: GitHub

- Email your GitHub username to `yfreund@ucsd.edu`
- You will be added to the course repo: **[repo URL]**
- All code, reports, and experiment logs live in the repo

Administration: WhatsApp

- Join the group: **[WhatsApp link]**
- WhatsApp: scheduling, quick questions
- GitHub: code, issues, decisions

Components of a learning system



- Training produces a model
- Testing runs the model on data it has not seen

What testing measures

- **Accuracy:** error rate on new data
- **Confidence:** does the model know when it may be wrong?
- **Speed:** time per prediction

Inference: throughput and latency

- **Latency:** time to answer one request
- **Throughput:** requests answered per second

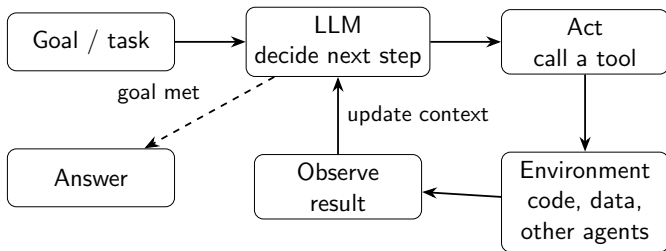


Plaud Note. Image: plaud.ai (editorial use)

- Pocket speech recognizers are too small for large speech models
- Audio is recorded on the device, then sent to the cloud for transcription
- Cost of the cloud:
 - network delay
 - needs connectivity
 - privacy
 - per-minute fees

The agent loop

- An agent is an LLM in a loop: read goal and context, decide, call a tool, observe, repeat
- **Tools:** run code, read/write files, search, call another agent
- **Memory:** the growing context of actions and observations
- In this course: learner agents train models; a coordinator agent runs the loop over them



The agent loop as a learning system

Agent loop

- LLM
- Tools
- Observation
- Goal

Learning system

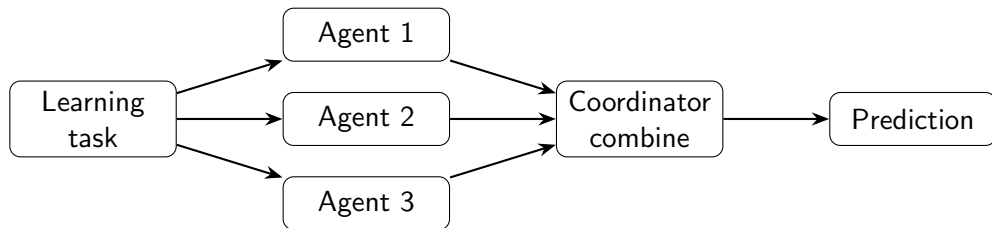
- Training: choose method and next update
- Calculate error, gradient, latency
- Error, gradient, and latency values
- Good performance on the test set

Caution

The agent sees only training and validation data; the test set is used once, at the end.

Ensembles of agents

- Sometimes it is useful to break the learning task into smaller learning tasks and combine the results
- Ensembles work that way
- Each small task goes to its own learner agent; a coordinator agent combines their predictions



Examples of ensembles

- **Bagging** (Breiman 1996): train many learners on bootstrap samples; majority vote
- **Random forests** (Breiman 2001): bagging of trees, each split chosen from a random subset of features
- **Boosting** (Freund and Schapire 1997): train learners in sequence, each focusing on the examples the previous ones got wrong; weighted vote
- **Decision trees!** Each split breaks the task into smaller learning tasks on subsets of the data; each leaf solves its own small task. Divide and conquer, rather than vote

Week-by-week plan

One student per week: present a paper (with slides), implement it or find an implementation, and run it on a Kaggle dataset. Report accuracy, confidence, and speed.

Week	Method	Paper	Presenter
2	Averaged perceptron	Freund and Schapire (1999)	Andrew Chen
3	Bagging	Breiman (1996)	Saisohan Shingade
4	Random forests	Breiman (2001)	Surya Setty
5	Boosting stumps	Freund and Schapire (1997)	Uday Lingampalli

Freund and Schapire (1999): “Large margin classification using the perceptron algorithm.” Students may swap weeks among themselves.